

HAONAN GE

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🏠 [Haonan-Ge.github.io](https://github.com/Haonan-Ge)

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RESEARCH INTERESTS

Large Language Model, multi-modal understanding, MLLM reasoning, Safety in MLLM, Agentic AI.

EDUCATION

Southeast University

Nanjing, China

B.Eng. in Electrical & Computer Engineering; GPA: 3.72/4.0

Sep 2022 – Expected Jun 2026

University of California, Irvine

Irvine, CA

Exchange in Computer Science; GPA: 4.0/4.0

Jun 2025 – Sep 2025

PUBLICATIONS

* denotes equal contribution

FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning

Haonan Ge, Yiwei Wang, Kai-Wei Chang, Hang Wu, Yujun Cai.

ICLR 2026, Under Review.

MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMS

Haonan Ge, Yiwei Wang, Ming-Hsuan Yang, Yujun Cai.

Findings of EMNLP 2025.

SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports

Haotian Xia*, Haonan Ge*, Junbo Zou*, Hyun Woo Choi, Xuebin Zhang, Danny Suradja, Botao Rui, Ethan Tran, Wendy Jin, Zhen Ye, Xiyang Lin, Christopher Lai, Shengjie Zhang, Junwen Miao, Shichao Chen, Rhys Tracy, Vicente Ordonez, Weining Shen, Hanjie Chen.

ICLR 2026, Under Review.

RefineShot: Rethinking Cinematography Understanding with Foundational Skill Evaluation

Hang Wu, Yujun Cai, Haonan Ge, Hongkai Chen, Ming-Hsuan Yang, Yiwei Wang.

ICLR 2026, Under Review.

RESEARCH EXPERIENCE

University of California, Merced & The University of Queensland

Merced, CA

Research Intern (Remote); Supervised by Prof. **Yujun Cai** & **Yiwei Wang**

May 2025 – Present

Project 1: FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning (lead)

- Proposed **Frame-Interleaved CoT (FiCOT)**: a multi-turn loop that alternates textual reasoning with active visual perception, enabling the model to *request* video evidence during reasoning (e.g., target frames or short clips).
- Introduced **Dynamic-Resolution Frame Sampling (DRFS)** to expose the policy to temporal-spatial trade-offs; developed **DRFS-GRPO** for group-relative optimization from outcome rewards (no frame-level labels).
- Implemented a compact **agent runtime**(prompt templates, tool schema, parsing/verification, 1–3 turn cap) and ran controlled ablations on short/long-video benchmarks (e.g., MVBench/MLVU/VideoMME).
- Demonstrated strong gains on long-video benchmarks (e.g., MLVU, VideoMME) with improved efficiency and interpretability.

Project 2: RefineShot: Rethinking Cinematography Understanding with Foundational Skill Evaluation

- Refined **ShotBench** with consistent option restructuring, releasing the expanded benchmark **RefineShot** for reliable evaluation.
- Performed the first critical analysis of **ShotVL** behavior, identifying issues in reasoning consistency and instruction adherence; proposed diagnostic checks.

- Designed an extended evaluation protocol to jointly assess task accuracy and foundational skills, improving fairness and reproducibility.

University of California, Irvine & Rice University **Irvine, CA & Houston, TX**
*Research Intern (On-site); Supervised by Prof. **Weining Shen & Hanjie Chen*** *Jun 2025 – Sep 2025*

Project: *SportR: A Benchmark for Multimodal LLM Reasoning in Sports* (**co-led**)

- Led **badminton** CoT annotations with progressive QA hierarchy; enforced quality control and consistency.
- Designed experiments for infraction/type/penalty/explanation/tactics and explicit **grounding** tasks.
- Implemented **SFT** pipeline and **GRPO**-based **RL** training (reward: LLM-as-Judge for QA; **GIoU** for grounding).
- Conducted RL ablations and runs, validating post-SFT → RL improvements and cross-modal generalization.
- Benchmarked open-source MLLMs (Qwen-VL, LLaVA-Next, GLM-4.5V, DeepSeek-VL) with unified prompts and judge ensemble.

The University of Queensland **Remote**
*Research Intern (Remote); Supervised by Prof. **Yujun Cai*** *Feb 2025 – May 2025*

Project: *MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMS*
 (EMNLP 2025 Findings; **first author**)

- Proposed a **training-free** decoding method that selects salient regions via cross-attention, generates per-region responses, and fuses them using **Jensen–Shannon Divergence**-based reliability weights.
- Designed **region-aware prompts** (CoT-inspired) and a **consistency-weighted fusion** procedure to improve factual grounding and reduce hallucinations without modifying model parameters.
- Built evaluation scripts for POPE/CHAIR/MME-Hallucination; observed consistent gains over strong decoding baselines (precision/F1 and CHAIR scores).

SERVICES

Reviewer: CVPR 2026, WACV 2026

SKILLS

Programming: Python, C/C++; Matlab; HTML/CSS/JS; $\text{\LaTeX}$
ML/Systems: PyTorch, Ray, FSDP; NumPy/Pandas; Git, Docker
OS: Linux (Ubuntu), Windows
Languages: Mandarin (Native), English (Fluent)

WORK EXPERIENCE

Comba Telecom (China) Ltd. **Guangzhou, China**
Software Engineer Intern *Jun 2023 – Aug 2023*

- Developed a communication framework using C# and TCP Sockets, implementing real-time frame transmission, data encapsulation, parsing, and log monitoring.
- Designed the user login interface and optimized the overall UI interaction logic to enhance system usability.

Southeast University, Volunteer Work **Nanjing, China**
Volunteer *Mar 2023 – Present*

- Completed 50+ hours of service organizing sports meets, admission interviews, and major campus events.

HONORS AND AWARDS

Outstanding Student Award **Nanjing, China**
Southeast University *2025*

Academic Competition Scholarship of Southeast University **Nanjing, China**
Awarded for outstanding achievements in academic competitions *2024*

The 15th National Mathematics Competition for College Students **Jiangsu, China**
First Prize *2023*

The 20th Jiangsu Provincial Higher Mathematics Competition **Jiangsu, China**
First Prize *2023*

REFERENCES

Prof. Kai-Wei Chang

Assistant Professor, University of California, Los Angeles

E-mail: kwchang.cs@gmail.com

Department: Computer Science

Prof. Yujun Cai

Assistant Professor, The University of Queensland

E-mail: yujun.cai@uq.edu.au

Department: Electrical Engineering and Computer Science

Prof. Yiwei Wang

Assistant Professor, University of California, Merced

E-mail: yiweiwang2@ucmerced.edu

Department: Computer Science, UC Merced NLP Lab

Prof. Weining Shen

Associate Professor, University of California, Irvine

E-mail: weinings@uci.edu

Department: Statistics